

Haiyi Wang

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Summary

A proficient Data Engineer with extensive experience in designing and maintaining data infrastructure, including overseeing critical data migrations and integration efforts at major corporations like VF Corporation and Carrefour. Renowned for enhancing efficiency in data pipelines and developing strategies to optimize loyalty programs, highlighted by a significant improvement in backend query performance by 20%. Eager to leverage strong technical skills and project experience to drive innovation and streamline processes in future Data Engineering endeavors.

Education

University of North Texas

Master of Computer Science, Computer Science

May 2023

- Achievements: GPA 4.0

East Normal of University Shanghai

Master of Finance, Finance

Shanghai University of Electric Power

Bachelor of Computer Science and Technology, Computer Science and Technology

Skills

Analysis Tools: Python (Pandas, NumPy, scikit-learn, matplotlib), R, SAS, Tableau, Power BI, advanced MS Excel

Machine Learning: Decision Tree, SVM, KNN, K-Means, XgBoost, Deep-Learning

Databases: Presto SQL, MySQL, Microsoft SQL Server, COSMOS, Oracle, MS Access

Big Data Technologies: Data Lake Storage, Hadoop, HDFS, MapReduce, Sqoop, Hive, Spark, Data Brick, Snowflake

Cloud: Azure, Google Cloud, AWS

Certification

AWS Certified Data Engineer - Associate

Issued May 2025 Expires May 2028

Credential ID 76e0539ac6ee42e1a534f97fbae3c9cb

<https://cp.certmetrics.com/amazon/en/public/verify/credential/76e0539ac6ee42e1a534f97fbae3c9cb>

Work Experience

VF Corporation

Data Engineer

Jul 2023 - Present

- Designed, built, and maintained data infrastructure and models that supported VF Corporation's loyalty programs, enhancing data accessibility and reliability
- Integrated data from various sources and monitored the performance of loyalty program initiatives using key performance indicators, improving enrollment and points redemption management
- Migrated data between OCP and Cheetah for user profile data and analyzed data discrepancies and quality, ensuring a smooth transition and data integrity
- Managed loyalty data up and downstream, ensuring alignment with reporting requirements for critical business drivers like OCP, OMS, CDP, and GDF, which improved data accuracy and reporting efficiency
- Drove transformation of loyalty program enablement through improved technology, processes, data management, and use case refinement for VF brands Vans, The North Face, and Timberland, resulting in enhanced program efficiency and customer satisfaction
- Developed and maintained loyalty dashboards and reports using Power BI to provide actionable insights into loyalty program metrics, enhancing decision-making processes
- Collaborated with product owner and brand controller to analyze customer behavior and loyalty program performance using SAS, leading to improved customer engagement strategies

Microsoft

Data Scientist

Seattle, WA

Oct 2022 - Apr 2023

- Wrote Scope scripts to extract user subscription and usage data from Cosmos DB, analyzing different features to enhance data insights

- Established a connection between Cosmos and Azure Data Lake Storage Gen1, transferring data to Data Lake to streamline data accessibility
- Utilized Data Bricks to process data, analyzing the relationship and relevance of features and subscriptions to improve data-driven decisions
- Applied Logistic Regression and Random Forest to build an ML model, identifying and targeting 'at risk' users to enhance user retention
- Deployed the ML model on Azure, help the Xbox marketing team to optimize and personalize campaigns to move members from high churn risk to low-risk segment. The retention improved 0.5%-1.2%.

University of North Texas

Dallas, TX

Graduate Teaching Assistant (Fundamental Database)

Feb 2022 - Aug 2022

- Administered exams, graded assignments, projects, and exams, and managed study sessions, resulting in improved student performance and understanding of database fundamentals
- Participated in the assessment process using various methods, such as quizzes and peer reviews, and provided effective, timely feedback, enhancing student engagement and learning outcomes
- Contributed to the development of teaching materials, including lecture slides and practice exercises, ensuring content and delivery met learning objectives, which led to increased student satisfaction and comprehension

Carrefour

Shanghai, China

Data Engineer

Jan 2016 - Dec 2017

- Identified possible options to address business problems in the supply chain e-commerce team, leading to improved decision-making processes
- Identified the most suitable data sources that aligned with business needs, enhancing data accuracy and reliability
- Analyzed complex data elements, dependencies, and relationships to contribute to physical/logical data models.
- Defined relational tables, primary/foreign keys, partition keys, and clusters in Google Big Query, optimizing data retrieval and storage
- Extracted data from Kafka topics, Google Cloud Platform, and other data sources, ensuring seamless data integration and availability
- Transformed and loaded data into relevant structures using appropriate techniques, improving data accessibility for analysis
- Developed solution and application features using Python, Scala, Spark, and SQL, enhancing system functionality and performance
- Created test cases to review and validate proposed solution designs, ensuring accuracy and reliability of implementations
- Deployed code to production servers and automated data pipelines, increasing deployment efficiency and reducing errors
- Supported business unit requirements and responded quickly to meet service level agreements, ensuring uninterrupted business operations
- Utilized code or table structure to improve the pipeline efficiency by 30%.
- Contributed code documentation, maintained playbooks, and provided timely progress updates.

Shanghai Lujiazui International Financial Asset Exchange Co.,Ltd

Shanghai, China

Data Analyst

Jul 2012 - Dec 2015

- Preprocessed and performed ETL for RTP team using Presto SQL, ensuring accurate extraction of testing data
- Prepared data pipeline using Airflow framework in Python and monitored pipeline jobs daily, ensuring smooth data flow and timely updates
- Migrated and extracted unstructured test logs from Manifold to MySQL, Hive, or Scuba, facilitating improved data accessibility and analysis
- Built dashboards in Tableau for stakeholders, enhancing visibility into server hardware validation performance
- Improved efficiency of data pipelines more than 30% and speed up backend query of dashboard by 50%.
- Built a Messenger Bot in Python to create and send out weekly customized test summary reports in the workplace group, improving team communication and efficiency

Projects

Natural Language Processing and Topic Modeling on User Review Dataset

- Applied unsupervised learning model and clustering customers into groups according to review texts by Python.
- Implemented tokenization and stemming technology to remove delimiters and stop words from each review text.
- Created term frequency - inverse document frequency (TF-IDF) matrix to weight factors and reflect the importance of a word in a corpus.
- Trained K-Means clustering model and Latent Dirichlet Allocation (LDA) to identify each review into different topic groups.

San Francisco Crime Analysis with Spark

- Used urllib to access and retrieve data from public websites.
- Performed descriptive analysis on SF crime datasets with Apache Spark RDD, PySpark and Spark SQL on data bricks.
- Generated story report based on crime trend and pattern and gave advice to tourists and distribution of police.